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We determine up to a logarithmic factor the order of magnitude of $R(3,K_t^r)$ for $r=3$:
\[c_1 t^{3/2}(\log t)^{-3/4} \leq R(3,K_t^3) \leq c_2 t^{3/2}\]
for some constants $c_1,c_2 > 0$. We also give bounds
on $R(3,K_t^r)$ for all $r \ge 4$ and related hypergraph Ramsey numbers.
The problem of determining the order of magnitude of $R(3,K_t^r)$ for each $r \geq 3$ remains open.

Hypergraph Ramsey Numbers: Triangles versus Cliques

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March 23, 2012

Abstract

A celebrated result in Ramsey Theory states that the order of magnitude of the graph Ramsey numbers $R(3,t)$ is $t^2/\log t$. In this paper, we consider an analogue of this problem for uniform hypergraphs. A *triangle* is a hypergraph consisting of edges e, f, g such that $|e\cap f| = |f\cap g| = |g\cap e| = 1$ and $e\cap f\cap g = \emptyset$. For all $r \geq 2$, let $R(3,K_t^r)$ be the smallest positive integer n such that in every red-blue coloring of the edges of the complete r-uniform hypergraph K_n^r, there exists a red triangle or a blue K_t^r. We determine up to a logarithmic factor the order of magnitude of $R(3,K_t^r)$ for $r = 3$:

$$c_1 t^{3/2}(\log t)^{-3/4} \leq R(3,K_t^3) \leq c_2 t^{3/2}$$

for some constants $c_1, c_2 > 0$. We also give bounds on $R(3,K_t^r)$ for all $r \geq 4$ and related hypergraph Ramsey numbers. The problem of determining the order of magnitude of $R(3,K_t^r)$ for each $r \geq 3$ remains open.

1 Introduction

Throughout this paper, C_k denotes a hypergraph *loose k-cycle*, namely the hypergraph with edges $e_1,\dots,e_k$ such that for $i \neq j$, $|e_i\cap e_j| = 1$ if $|i-j| = 1 \mod k$ and $e_i \cap e_j = \emptyset$ otherwise. In particular, a *loose triangle* is a hypergraph consisting of three edges e, f, g such that $|e\cap f| = |f\cap g| = |g\cap e| = 1$ and $e\cap f\cap g = \emptyset$. Since we consider only loose cycles and triangles, we will omit the word "loose". An *r-graph* is an r-uniform hypergraph. An *independent set* in a hypergraph is a set of vertices containing no edges of the hypergraph. Let K_t^r denote the t-vertex *complete r-graph*, i.e., the t-vertex r-graph whose edges are all r-element subsets of $[t]$. In this paper we consider the cycle versus complete hypergraph Ramsey numbers $R(C_k, K_t^r)$ – this is the minimum n such that every n-vertex r-graph contains either a cycle C_k or an independent set of t vertices. Our main effort will be on the triangle - complete hypergraph Ramsey number $R(C_3, K_t^r)$. A celebrated result of Kim [10] together with earlier bounds by Ajtai, Komlós and Szemerédi [2] shows

$$R(C_3, K_t) = \Theta\Big(\frac{t^2}{\log t}\Big) \quad \text{as } t \to \infty.$$

This establishes the order of magnitude of these Ramsey numbers for graphs.

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1.1 Hypergraph Triangles

Motivated by the triangle-complete graph Ramsey numbers, in this paper we determine the order of magnitude of the triangle-complete Ramsey numbers for triple systems up to logarithmic factors:

Theorem 1. *There exist constants $c_1, c_2 > 0$ such that for all $t \geq 1$,*

$$\frac{c_1 t^{3/2}}{(\log t)^{3/4}} \leq R(C_3, K_t^3) \leq c_2 t^{3/2}.$$

The upper bound in Theorem 1 is proved in Section 2. The lower bound in Theorem 1, which comes from a construction that combines randomness and linear algebra, is in Section 3. It is convenient to refer to this construction as a *random block construction*. Some of the ideas of the random block construction were recently used in [11] to study a related problem. We shall see that the random block construction for Theorem 1 extends in a straightforward manner to give lower bounds for $R(C_3, K_t^r)$ for all $r \geq 3$:

Theorem 2. *Let $r \geq 3$. Then for some constant $c > 0$ and all $t \geq 1$,*

$$c\Big(\frac{t}{\log t}\Big)^{3/2} \leq R(C_3, K_t^r) \leq (t+1)^2.$$

The proof of the upper bound in Section 2 shows more generally that any (not necessarily uniform) triangle-free n-vertex hypergraph without singleton edges has an independent set of size at least $\lfloor \sqrt{n} \rfloor$. The proof of the lower bound is given in Section 5. In light of Theorem 1, we make the following conjecture:

Conjecture 1.

$$R(C_3, K_t^3) = o(t^{3/2}) \qquad \textit{as } t \to \infty.$$

It would also be interesting to determine whether for all $r \geq 2$, $R(C_3, K_t^r) = o(t^2)$ as $t \to \infty$.

1.2 Arithmetic progressions

A motivation for studying triangle-complete hypergraph Ramsey numbers is the notorious extremal problem for three-term arithmetic progressions. Let $r_3(N)$ denote the largest size of a set of integers in $\{1, 2, \dots, N\}$ containing no three-term arithmetic progressions. This problem has attracted much attention, starting with the original theorems of van de Waerden and Roth. The best known bounds are as follows: for some constant $c > 0$,

$$\frac{N}{e^{c\sqrt{\log N}}} \leq r_3(N) \leq \frac{N}{(\log N)^{1-o(1)}}.$$

The lower bound, which comes from a construction of Behrend [3], is essentially unchanged for more than sixty years. The upper bound, due to Sanders [16] improves many earlier results which gave smaller powers of $\log N$ in the denominator. Let $RL(C_3, K_t^3)$ denote the minimum n such that every n-vertex *linear* triangle-free r-graph has an independent set of size t. Then $RL(C_3, K_t^3) \leq R(C_3, K_t^3) \leq c_2 t^{3/2}$ by Theorem 1. We prove the following:

Theorem 3. *For some constants $c_1, c_2 > 0$,*

$$\frac{t^{3/2}}{e^{c_1\sqrt{\log t}}} \leq RL(C_3, K_t^3) \leq c_2 \frac{t^{3/2}}{(\log t)^{1/2}}.$$

The upper bound is proved in Section 2, and the lower bound is given in Section 5. This theorem is perhaps some support for believing $R(C_3, K_t^3) = o(t^{3/2})$. This relates to upper bounds on $r_3(N)$ as follows: if one is able to show

$$RL(C_3, K_t^3) = O\left(\frac{t^{3/2}}{(\log t)^{c+3/4}}\right)$$

for some $c > 0$, then we shall see that

$$r_3(N) = O\left(\frac{N}{(\log N)^{3c}}\right).$$

This offers perhaps a new approach to estimating $r_3(N)$, and is explained at the end of Section 5.

1.3 Hypergraph cycles

The random block construction for Theorem 1 extends more generally to give lower bounds on all *cycle-complete hypergraph Ramsey numbers*. The cycle C_3 is precisely a hypergraph triangle. We give for all $k, r \geq 3$ a construction of C_k-free r-graphs with low independence number, based on known results on C_k-free bipartite Ramanujan graphs of Lubotzky, Phillips and Sarnak [13]. Specifically, we prove the following theorem by a suitable and fairly straightforward modification of the random block construction. We write $f = O^*(g)$ to denote that for some constant c, $f(t) = O((\log t)^c g(t))$, and $f = \Omega^*(g)$ denotes that $g = O^*(f)$.

Theorem 4. *For $r, k \geq 3$,*

$$R(C_k, K_t^r) \quad = \quad \Omega^*\left(t^{1+\frac{1}{3k-1}}\right) \qquad \textit{as } t \to \infty.$$

The key point of this theorem is that the exponent $1 + 1/(3k-1)$ of t is bounded away from 1 by a constant independent of r, and strictly improves for all $r, k \geq 3$ the lower bounds given by considering appropriate random hypergraphs, namely

$$R(C_k, K_t^r) = \Omega^*\left(t^{1+\frac{1}{kr-k-r}}\right) \qquad \text{as } t \to \infty.$$

In fact, using more complicated results about the distribution of prime numbers, we can improve the exponent $1 + 1/(3k-1)$ slightly.

In the case $r = 2$, for graphs, the best available constructions for lower bounds on $r(C_k, K_t^r)$ indeed come from appropriate random graphs; in particular the C_k-free random graph process studied by Bohman and Keevash [6]. By using the known constructions of extremal bipartite graphs of girth 12, arising from generalized hexagons, the random block construction offers the following improvement of Theorem 4 for pentagons:

Theorem 5. *For $r \geq 3$,*

$$R(C_5, K_t^r) \quad = \quad \Omega^*(t^{5/4}) \qquad \text{as } t \to \infty.$$

We suspect the exponent 5/4 above may be tight, and perhaps even more generally, $r(C_k, K_t^r) = \Theta^*(t^{k/(k-1)})$ for all $r, k \geq 3$. The proofs of Theorems 4 and 5 are outlined in Section 5 by indicating the requisite modifications to the proof of the lower bound in Theorem 1.

2 Proof of Theorems 1 – 3 : Upper Bounds

The goal of this section is to prove the upper bounds in Theorems 1, 2 and 3. We first give some general lower bounds on the independence number of triangle-free hypergraphs. Recall the *chromatic number* $\chi(H)$ of a hypergraph H is the minimum k such that there is an assignment of k colors to the vertices such that no subset of vertices of the same color forms an edge of H. The upper bound in Theorem 2 follows directly from the following:

Theorem 6. *Let H be any hypergraph on n vertices not containing a triangle and in which $|e| \geq 2$ for all $e \in H$. Then*

$$\alpha(H) \geq \lfloor \sqrt{n} \rfloor.$$

Proof. Suppose for a contradiction that $\alpha(H) < \lfloor \sqrt{n} \rfloor$. Then $\chi(H) > k := \lfloor \sqrt{n} \rfloor$. So, H contains a $(k+1)$-vertex-critical subgraph H', which means that $\chi(H') = k+1$ but $\chi(H' - v) \leq k$ for every $v \in V(H')$. By Corollary 3 on Page 431 of [5] (see also [18] and [12]), the *strong degree* of each vertex in H' is at least k, i.e. for each $v \in V(H')$ there are k edges $e_1, e_2, \ldots, e_k$ such that $e_i \cap e_j = \{v\}$ for all $1 \leq i < j \leq k$. In words, the e_is share v and nothing else. Choose a vertex v_i in each $e_i \backslash \{v\}$. Since H' has no triangles, the set $\{v_1, \ldots, v_k\}$ is an independent set of H of size $k \geq \lfloor \sqrt{n} \rfloor$, which is a contradiction. □

The proof above actually implies that if H has no edges of size two, and $\alpha(H) < \sqrt{n}$, then H contains not only a triangle, but a configuration with more edges that contains a triangle. A second result that will be useful to us is the well-known lower bound on the independence number of an n-vertex r-graph of average degree d. We include a proof for completeness:

Proposition 7. *Let H be an n-vertex r-graph of average degree d, where $r \geq 2$. Then*

$$\alpha(H) \geq \frac{(r-1)n}{rd^{1/(r-1)}}.$$

If in addition H is linear, then for fixed r,

$$\alpha(H) = \Omega\Big(\frac{n(\log d)^{1/(r-1)}}{d^{1/(r-1)}}\Big).$$

Proof. The second statement is from Duke, Lefmann and Rödl [7], building on an earlier theorem of Ajtai, Komlós, Pintz, Spencer and Szemerédi [1]. For the first statement, randomly pick vertices of H independently with probability p. Let X be the set of picked vertices, and let Y be the set of picked edges – edges all of whose vertices are picked. Then

$$E(|X| - |Y|) = pn - p^r|H| = pn - \frac{p^r dn}{r}.$$

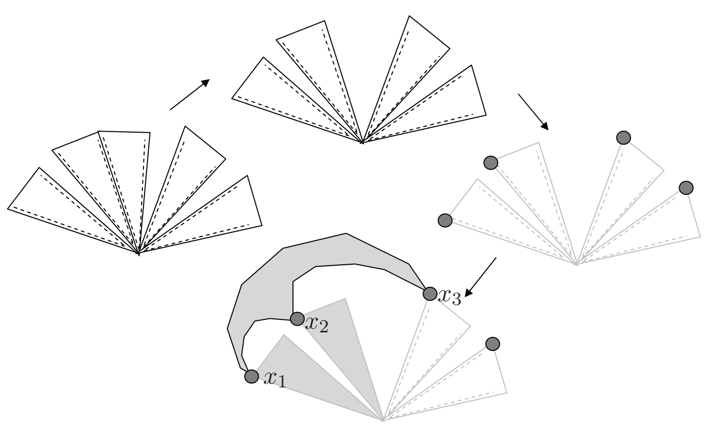
Also, $\alpha(H) \geq E(|X| - |Y|)$, since an independent set of H of size $|X| - |Y|$ is obtained by deleting from X one picked vertex from every picked edge. Taking $p = d^{-1/(r-1)}$ gives the result. $\square$

2.1 Proof of Theorem 1 : Upper Bound

In this section, we intend to prove the upper bound in Theorem 1 by showing that an n-vertex C_3-free triple system contains an independent set of size at least $n^{2/3}/10$. Let H be a triangle-free triple system on n vertices. If $|H| \leq n^{5/3}$, then $\alpha(H) \geq (3-1)n^{2/3}/3 \geq n^{2/3}/10$, by Proposition 7. We can assume henceforth that $|H| > n^{5/3}$. An edge $e \in H$ is called *k-light* if exactly k pairs of vertices of e have codegree at most four. An edge is *heavy* if it is 0-light: we see quickly that a triangle-free triple system has no heavy edges: for a heavy edge $\{x, y, z\} \in H$, we can easily (for example, greedily) pick distinct vertices $a, b, c \notin \{x, y, z\}$ such that $\{a, x, y\}, \{b, y, z\}, \{c, x, z\}$ is a triangle, since each pair has codegree more than four. We now consider two cases.

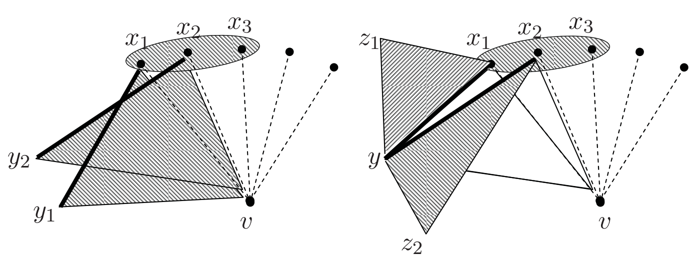
Case 1. *The number of edges in H that are 2-light or 3-light is at least $|H|/2$.*

In this case, some vertex v of H is in at least $|H|/2n$ edges, where two pairs of codegree at most four in each edge contain v. Let $e_1, e_2, \ldots, e_k$ be such a set of edges on v. Then the link graph $L(v)$ consisting of pairs $e_i \backslash \{v\}$ has maximum degree at most four. It follows by Vizing's Theorem that $L(v)$ has a matching of size $\ell \geq k/5$. This means we have found edges, say $e_1, e_2, \ldots, e_\ell$ sharing no vertices other than v, and such that two pairs in each edge have codegree at most four and both those pairs contain v. Now pick $x_1, x_2, \ldots, x_\ell$ where $x_i \in e_i \backslash \{v\}$ for $1 \leq i \leq \ell$. We claim this is an independent set. If not, then say $e = \{x_1, x_2, x_3\} \in H$. Then $\{e, e_1, e_2\}$ is a triangle in H, since e_1, e_2 only share v, e and e_1 only share x_1, and e and e_2 share only x_2 – see the figure. This independent set has size $\ell \geq k/5 \geq |H|/10n$. This completes the proof in Case 1.



Case 2. *The number of 1-light edges in H is at least $|H|/2$.*

By averaging, some v in H lies in $|H|/n$ 1-light edges and the pair of codegree at most four in each edge contains v. Choose edges $e_1, e_2, \ldots, e_k$ containing v containing distinct pairs $\{v, x_1\}, \ldots, \{v, x_k\}$ of codegree at most four, where $k \geq |H|/5n$. This bound on k is possible since each pair $\{v, x\}$ of codegree at most four accounts for at most four edges containing v that also contain x. Note also that there are exactly two pairs in each of the edges e_i that have codegree more than four. We use this fact to prove that $\{x_1, x_2, \ldots, x_k\}$ is again an independent set. Suppose not, and that $\{x_1, x_2, x_3\}$ is an edge. Let $e_i \backslash \{x_i, v\} = \{y_i\}$. Note that every y_i is disjoint from $\{x_1, x_2, x_3\}$, otherwise if say $y_i = x_j$, then $\{v, x_i\}$ and $\{v, x_j\}$ both have codegree at most 4, but they lie in the edge e_i, which only has one pair of codegree at most four – a contradiction. So every y_i is disjoint from $\{x_1, x_2, x_3\}$. Now we claim $y_1 = y_2 = y_3$. If say $y_1 \neq y_2$ (left figure below), consider the triples $\{v, x_1, y_1\}, \{v, x_2, y_2\}$ and $\{x_1, x_2, x_3\}$. Since y_1, y_2, x_3 are all distinct, this is a triangle. So $y_1 = y_2 = y_3 = y$. Now consider the pairs $\{y, x_1\}, \{y, x_2\}$ (shown in black bold lines in the right figure below). Since $\{y, x_1\}$ and $\{y, x_2\}$ are respectively pairs in e_1 and e_2 and they do not contain v, by choice of e_1 and e_2 those pairs have codegree more than four. So we can pick $z_1 \neq z_2$ with $z_1, z_2 \notin \{x_1, x_2, x_3, y_1, y_2, y_3, v\}$ such that $\{x_1, y, z_1\}, \{x_2, y, z_2\}, \{x_1, x_2, x_3\}$ is a triangle – namely z_1, z_2, x_3 are all distinct. This shows $\{x_1, x_2, \ldots, x_k\}$ is independent, and it has size $k \geq |H|/5n > n^{2/3}/10$. $\square$



2.2 Proof of Theorem 3 : Upper Bound

To prove Theorem 3, it is sufficient to show that every n-vertex triangle-free linear triple system has an independent set of size $\Omega(n^{2/3}(\log n)^{1/3})$. Let H be such a triple system. By [7] (see Proposition 7),

$$\alpha(H) = \Omega\Big(\frac{n\sqrt{\log d}}{\sqrt{d}}\Big)$$

where d is the average degree of H. The union of all pairs $e \backslash \{v\}$ for edges e containing a vertex v of degree at least d is an independent set of $2d$ vertices in H, since H is linear and triangle-free.

Therefore

$$\alpha(H) = \Omega\Big(\min\Big\{d, \frac{n\sqrt{\log d}}{\sqrt{d}}\Big\}\Big) = \Omega(n^{2/3}(\log n)^{1/3}).$$

This completes the proof of the upper bound in Theorem 3.

3 Proof of Theorem 1 : Lower Bound

Generalized quadrangles were first constructed by Tits [17] and described as graphs by Benson [4]. Let G_q denote a *generalized quadrangle of order* q, which is a $(q+1)$-regular $(q+1)$-uniform linear triangle-free hypergraph on q^3+q^2+q+1 vertices. Generalized quadrangles of order q exist whenever q is a prime power. Based on G_q, we now specify the construction of a triangle-free n-vertex hypergraph H_q with independence number at most $400n^{2/3}\sqrt{\log n}$ for large enough n, which gives the lower bound in Theorem 1.

Let $\tau = \lfloor 200\sqrt{\log q}\rfloor$. To describe H_q, we require an auxiliary hypergraph F_q with vertex set $[q+1]$, defined as follows. Let $V = \{v_{ij} : 1 \le i, j \le \tau\}$ be a τ^2-element subset of $[q+1]$ and let $S_1, \dots, S_\tau, T_1, \dots, T_\tau$ be a partition of $[q+1] - V$ into sets whose sizes differ by at most one. Let $S = \bigcup_{i=1}^{\tau} S_i$ and $T = \bigcup_{j=1}^{\tau} T_j$. The edge set of F_q is the set of all triples $\{v_{ij}, a, b\}$ such that $a \in S_i$ and $b \in T_j$. Note that F_q is actually 3-partite, with parts V, S and T. Then H_q is constructed by taking independently for each $e \in G_q$ a random bijection π_e from $V(F_q)$ to e and letting a triple in e be an edge if its pre-image is an edge in F_q.

Lemma 8. *H_q is triangle-free.*

Proof. Since G_q is linear and triangle-free, it is sufficient to verify that F_q is triangle-free. Suppose F_q has a triangle. Then two of the triples contain a vertex $v_{ij} \in V$, say these triples are $\{v_{ij}, s_i, t_j\}$, $\{v_{ij}, s_i', t_j'\}$. Now the third triple must be $\{v, s_i, t_j'\}$ or $\{v, s_i', t_j\}$ for some $v \in V$. By definition of F_q, this implies $v = v_{ij}$, a contradiction. □

Next we bound from above the probability that a set of τ vertices of $e \in E(G_q)$ is an independent set in H_q.

Lemma 9. *Let I be a τ-element subset of $e \in E(G_q)$. If q is large enough, then the probability that I is independent in H_q is at most $1 - \frac{\tau^3}{11q}$.*

Proof. Let N be the number of τ-sets X of $V(F_q) = [q+1]$ that are not independent in F_q. A lower bound for N is obtained by picking an element $v_{ij} \in V$, an element $s \in S_i$, an element $t \in T_j$

and $\tau-3$ elements in $[q+1]-(V\cup S_i\cup T_j)$. As $q\to\infty$, this gives

$$\begin{aligned}
N &\geq \sum_{v_{ij}\in V} |S_i||T_j|\binom{q+1-|S_i|-|T_j|-|V|}{\tau-3} && (1)\\
&\sim \tau^2\left(\frac{q}{2\tau}\right)^2\binom{(1-1/\tau)q}{\tau-3} && (2)\\
&\sim \frac{q^2}{4}(1-1/\tau)^{\tau-3}\frac{q^{\tau-3}}{(\tau-3)!} && (3)\\
&\sim \frac{\tau^3}{4eq}\frac{(q+1)^t}{\tau!} && (4)\\
&\sim \frac{\tau^3}{4eq}\binom{q+1}{\tau}. && (5)
\end{aligned}$$

Consequently, for large enough q, the number of τ-sets of $V(F_q)$ that are independent in F_q is at most $(1-\frac{\tau^3}{11q})\binom{q+1}{\tau}$. Now the probability that $I\subset e$ is not independent in H_q is

$$\frac{|\{\pi_e:\ I \text{ is not independent under } \pi_e\}|}{(q+1)!}=\frac{N\tau!(q+1-\tau)!}{(q+1)!}=\frac{N}{\binom{q+1}{\tau}}.$$

The lower bound on N now gives the desired result. $\square$

3.1 A general spectral lemma

In this section, we employ a lemma which relates the distribution of edges in a bipartite graph to spectral properties of its adjacency matrix. This lemma is an analog of a well-known spectral lemma in graph theory which is frequently referred to as the expander mixing lemma, and is used especially in the context of (n,d,λ)-graphs and pseudorandom graphs. The lemma we give, which may be referred to as the expander mixing lemma for bipartite graphs, appears in a different form in [8] and in [9]. For completeness, we give the proof here.

Lemma 10. *Let $G(U,V)$ be a d-regular bipartite graph with adjacency matrix A and let $\lambda_1\geq\lambda_2\geq\cdots\geq\lambda_N$ be the eigenvalues of A. Let $\lambda=\max\{|\lambda_i|: i\notin\{1,N\}\}$. Then for any sets $X\subseteq U$ and $Y\subseteq V$, the number $e(X,Y)$ of edges from X to Y satisfies*

$$\left|e(X,Y)-\frac{d}{|V|}|X||Y|\right| \;\leq\; \lambda\sqrt{|X||Y|}.$$

Proof. Let χ_X and χ_Y denote the characteristic vectors of X and Y. Let $x_1,x_2,\ldots,x_N$ be an orthonormal basis of eigenvectors of A, where x_i is the eigenvector corresponding to λ_i, and write

$$\chi_X=\sum_{i=1}^N s_ix_i \quad \chi_Y=\sum_{i=1}^N t_ix_i.$$

Then

$$e(X,Y)=\langle A\chi_X,\chi_Y\rangle=\lambda_1s_1t_1+\lambda_Ns_Nt_N+\sum_{i=2}^{N-1}\lambda_is_it_i.$$

The values of s_1, t_1, s_N and t_N are recovered quickly from the knowledge of the first and last eigenvectors, x_1 and x_N, recalling x_1 is the constant unit vector and x_N is the unit vector which is constant on $V(G_q)$ and minus that constant on $E(G_q)$. Noting that $\|\chi_X\|^2 = |X|$ and $\|\chi_Y\|^2 = |Y|$, and using $\lambda_1 = d = -\lambda_N$, it is straightforward to see

$$e(X,Y) = \frac{d}{|V|}|X||Y| + \sum_{i=2}^{N-1} \lambda_i s_i t_i.$$

Finally, by Cauchy-Schwarz,

$$\sum_{i=2}^{N-1} \lambda_i s_i t_i \leq \lambda(A) \left(\sum_{i=1}^{N} s_i^2\right)^{1/2} \left(\sum_{i=1}^{N} t_i^2\right)^{1/2}$$

and the sums are $\|\chi_X\| = \sqrt{|X|}$ and $\|\chi_Y\| = \sqrt{|Y|}$ respectively. $\square$

This lemma will be used in the context of hypergraphs (in particular for the hypergraph G_q) in the following way: if H is a hypergraph, then the *bipartite incidence graph* of H is the bipartite graph $B(H)$ whose parts are $V(H)$ and $E(H)$, and $\{v, e\} \in E(B(H))$ if and only if $v \in e$. We denote by $A(H)$ the adjacency matrix of the bipartite incidence graph $B(H)$, and when $|V(H)| = |E(H)|$ we denote by $\lambda(H)$ the largest absolute value of the eigenvalues of $A(H)$ other than λ_1 and λ_N. Lemma 10 is applied to $B(H)$ to give the following hypergraph formulation:

Lemma 11. *Let H be a d-uniform d-regular hypergraph and let $X \subseteq V(H)$ and $Y \subseteq E(H)$. Then*

$$\Big|\sum_{e \in Y} |X \cap e| - \frac{d}{|V|}|X||Y|\Big| \;\leq\; \lambda(H)\sqrt{|X||Y|}.$$

In particular, if $\lambda(H) \leq \delta\sqrt{d}$ and $|X| \geq 2\tau n/d$, then the number of edges $e \in E(G_q)$ such that $|X \cap e| \geq \tau$ is at least $n - 2\delta^2 n/\tau$.

Proof. For the first inequality, if H is a d-uniform d-regular hypergraph, then $B(H)$ is d-regular. Applying Lemma 10 gives

$$\Big|e(X,Y) - \frac{d}{|V|}|X||Y|\Big| \;\leq\; \lambda(H)\sqrt{|X||Y|}.$$

We note that

$$e(X,Y) = \sum_{e \in Y} |X \cap e|.$$

This gives the first inequality of Lemma 11. Applying this inequality with $\lambda(G_q) \leq \delta\sqrt{d}$, we obtain for any $Z \subseteq E(G_q)$,

$$\Big|\sum_{e \in Z} |X \cap e| - \frac{(q+1)}{n}|X||Z|\Big| \leq \delta\sqrt{d|X||Z|}.$$

Now let $Y = \{e \in E(G_q) : |X \cap e| \geq \tau\}$ and $Z = E(G_q)\backslash Y$. Suppose for a contradiction that $|Z| > 2\delta^2 n/\tau$. By definition of Z,

$$\sum_{e\in Z} |X \cap e| < \tau|Z|.$$

By the preceding inequality,

$$\tau|Z| > \sum_{e\in Z} |X \cap e| > \frac{d}{n}|X||Z| - \delta\sqrt{d|X||Z|}.$$

Since $|X| \geq 2\tau n/d$, we get

$$\tau|Z| < \delta\sqrt{2\tau n|Z|}.$$

This contradicts $|Z| > 2\delta^2 n/\tau$. $\square$

We remark that for fixed $|X|$, $d|X|/|V|$ is exactly the expected value of $|X \cap e|$ when X is a random set whose elements are chosen from $V(H)$ independently with probability $|X|/|V|$.

3.2 Spectral properties of $A(G_q)$

In order to apply Lemma 11 to G_q, we determine $\lambda(G_q)$. Since G_q is $(q+1)$-uniform and $(q+1)$-regular, the bipartite incidence graph $B(G_q)$ is $(q+1)$-regular. Since $B(G_q)$ is connected, this implies $q+1$ and $-(q+1)$ are eigenvalues of $A(G_q)$ with multiplicity 1. To determine $\lambda(A)$, we use the well known property of G_q that

$$A^3 = J + qA$$

where J is the block matrix

$$J = \begin{pmatrix} 0 & K^t \\ K & 0 \end{pmatrix}.$$

and K is the square all 1 matrix with appropriate dimensions. If $\lambda \notin \{-(q+1), q+1\}$ is an eigenvalue of A, then an eigenvector x for λ is orthogonal to the constant unit vector and so $Kx = 0$. It follows that $\lambda^3 = q\lambda$ and therefore $\lambda \in \{-\sqrt{q}, 0, \sqrt{q}\}$. Since the eigenvalues of $A(G_q)$ other than $-(q+1)$ and $(q+1)$ are not all zero, $\lambda(G_q) = \sqrt{q}$. A more complete analysis of these eigenvalues and their multiplicities was achieved by Haemers [8]. Since we have $\lambda(G_q) = \sqrt{q}$, Lemma 11 gives the following:

Corollary 12. *Let $X \subseteq V(G_q)$ where $|X| \geq 2\tau n/(q+1)$, and let Y be the set of $e \in E(G_q)$ such that $|X \cap e| \geq \tau$. Then*

$$|Y| \geq n - \frac{2n}{\tau}.$$

Proof. Since $\lambda(G_q) = \sqrt{q}$, applying Lemma 11 with $\delta = 1$ and $d = q+1$ gives the result. $\square$

3.3 Independence number of H_q

To finish the proof of the lower bound in Theorem 1, we have to show for all n there exists a triangle-free triple system on n vertices with independence number $O(n^{2/3}\sqrt{\log n})$ as $n \to \infty$. If $n = q^3 + q^2 + q + 1$ for some prime power q, then we show that with positive probability, H_q has no independent set of size more than $2\tau n/q$ if n is large enough and $\tau = \lfloor 200\sqrt{\log q}\rfloor$. Note that $2\tau n/q = O(n^{2/3}\sqrt{\log n})$. If n is not of this form, pick the smallest prime power q such that $n \le q^3 + q^2 + q + 1$, and remove $q^3 + q^2 + q + 1 - n$ vertices from H_q. The new hypergraph H'_q has $\alpha(H'_q) \le \alpha(H_q)$. Since it is well-known that there exists a prime $q : n^{1/3} \le q \le 2n^{1/3}$, H'_q has no independent set of size more than $2\tau n/q = O(n^{2/3}\sqrt{\log n})$, as required.

Suppose $X \subset V(H_q) = V(G_q)$ is an independent set of size $2\tau n/q$ in H_q. By Corollary 12, at least $n - 2n/\tau$ of the edges of G_q contain at least τ vertices of I. Let Y be this set of edges. For each $e \in Y$, $X \cap e$ is an independent set in the random hypergraph F_q on e. Let B_e be the event that $X \cap e$ is independent in F_q. By Lemma 9,

$$P(B_e) \;\; \le \;\; 1 - \frac{\tau^3}{11q}$$

provided q is large enough. The events B_e are independent over $e \in Y$, and therefore the expected number of independent sets of size $2\tau n/q$ in H_q is at most

$$\begin{aligned} \prod_{e \in Y} P(B_e) \;\; &\le \;\; \Big(1 - \frac{\tau^3}{11q}\Big)^{n-2n/\tau}\binom{n}{2\tau n/q} \\ &\le \;\; \exp\Big(-\frac{n\tau^3}{20q} + \frac{2\tau n \log n}{q}\Big) \end{aligned}$$

provided q is large enough. Since $\tau = \lfloor 200\sqrt{\log q}\rfloor$, the above quantity decays to zero as $q \to \infty$, recalling $n = q^3 + q^2 + q + 1$. Therefore with high probability, H_q has no independent set of size more than $2\tau n/q < 400n^{2/3}\sqrt{\log n}$. This proves Theorem 1. $\square$

4 Proofs of Theorems 2 – 5

The proofs of the lower bounds in Theorems 2 – 5 are all based on slight modifications of the construction of H_q and F_q presented in the last section. We describe the required modifications for each theorem.

4.1 Proof of Theorem 2

For Theorem 2, which states for all $r \ge 3$ that

$$R(C_3, K_t^r) = \Omega\Big(\frac{t^{3/2}}{(\log t)^{3/2}}\Big),$$

we take $H_{q,r}$ to consist of copies of a hypergraph $F_{q,r}$, where for each edge e of G_q, we pick a random bijection $\pi_e : [q+1] \to e$ and place a copy of $F_{q,r}$ in e using π_e. It is sufficient to show that $\alpha(H_{q,r}) = O(n^{2/3}\log n)$ when $n = q^3+q^2+q+1$ for some prime power q, and to use the distribution of primes as in the proof of Theorem 1 to complete the proof for $n \neq q^3+q^2+q+1$.

The hypergraph $F_{q,r}$ is the r-graph on $q+1$ vertices defined as follows. Let $\tau = \lfloor r!30\log q\rfloor$. Let $V = \{v_1,\dots,v_\tau\} \subset [q+1]$. Let $S_1,\dots,S_\tau$ be a partition of $[q+1]-V$ where S_i's differ in size by at most one. The edge set of $F_{q,r}$ is the collection of r-sets $\{v_i,a_1,\dots,a_{r-1}\}$ such that $a_j \in S_i$ for all $j = 1,\dots,r-1$. In other words, $F_{q,r}$ is a disjoint union of *stars*. The key lemma about $F_{q,r}$ is now as follows:

Lemma 13. *Let $r \geq 3$ and I be a τ-element subset of $e \in E(G_q)$, where $\tau = \lfloor r!30\log q\rfloor$. If q is large enough, then the probability that I is independent in $H_{q,r}$ is at most $1-\frac{\tau^2}{r!q}$.*

Proof. Let N be the number of τ-sets X of $V(F_{q,r}) = [q+1]$ that are not independent in $F_{q,r}$. A lower bound for N is obtained by picking an element $v_i \in V$, $r-1$ elements in S_i, and $\tau - r$ elements in $[q+1]-(V\cup S_i)$. As $q\to\infty$, this gives

$$N \geq \sum_{v_i\in V}\binom{|S_i|}{r-1}\binom{q+1-|S_i|-|V|}{\tau-r} \sim \frac{\tau^2}{e(r-1)!q}\binom{q+1}{\tau}.$$

Consequently, for large enough q, the number of τ-sets of $V(F_{q,r})$ that are independent in $F_{q,r}$ is at most $(1-\frac{\tau^2}{r!q})\binom{q+1}{\tau}$. Now the probability that $I\subset e$ is not independent in $H_{q,r}$ is

$$\frac{|\{\pi_e:\ I \text{ is not independent under } \pi_e\}|}{(q+1)!} = \frac{N\tau!(q+1-\tau)!}{(q+1)!} = \frac{N}{\binom{q+1}{\tau}}.$$

The lower bound on N now gives the desired result. $\square$

The rest of the proof for $H_{q,r}$ carries through as for H_q, except at the end, the expected number of independent sets of size $2\tau n/q$ in $H_{q,r}$ is now by Corollary 12 at most

$$\Big(1-\frac{\tau^2}{r!q}\Big)^{n-2n/\tau}\binom{n}{2\tau n/q} < \exp\Big(-\frac{\tau^2 n}{r!2q}+\frac{2\tau n\log n}{q}\Big).$$

The choice of $\tau = \lfloor r!30\log q\rfloor$ ensures this decays to zero. We conclude that

$$\alpha(H_{q,r}) = O(n^{2/3}\log n)$$

and this gives the lower bound on Ramsey numbers in Theorem 2.

4.2 Proof of Theorem 3

Based on the hypergraph G_q, for $n = q^3+q^2+q+1$ and q a prime power, we construct an n-vertex linear triangle-free hypergraph H_q^* with $\alpha(H_q^*) < n^{2/3}/\exp(c\sqrt{\log n})$ for some $c > 0$. If n is not of that form, then as in the proof of Theorem 1 we use the distribution of primes and a large subhypergraph of H_q^* to obtain the same result with perhaps a slightly larger constant c.

Let F_q^* be the hypergraph on $q+1$ vertices defined as follows. Put $m = \lfloor (q+1)/6 \rfloor$ and let $A \subset \{1, 2, \ldots, m\}$ be a set containing no three-term arithmetic progressions. To define F_q^*, let

$$V(F_q^*) = \{x_1, x_2, \ldots, x_m, y_1, y_2, \ldots, y_{2m}, z_1, z_2, \ldots, z_{3m}, w_{3m+1}, \ldots, w_{q+1}\}$$

and let

$$E(F_q^*) = \{\{x_i, y_{i+a}, z_{i+2a}\} : 1 \leq i \leq m, a \in A\}.$$

It is straightforward to see that F_q^* is linear. Ruzsa and Szemerédi [15] observed that F_q^* is triangle-free due to the fact that A contains no 3-term progression. Note also $|F_q^*| = m|A|$. The main lemma we require counts independent sets of size τ in F_q^*.

Lemma 14. *If q is large enough, $\tau \geq 50$ and $|A|\tau^2 \leq q$, then the number of independent sets of size τ in F_q^* is at most*

$$\Big(1 - \frac{|A| \cdot \tau^3}{10q^2}\Big)\binom{q+1}{\tau}.$$

Proof. Let N be the number of non-independent sets of size τ in F_q^*. It is sufficient to show

$$N \geq \frac{|A| \cdot \tau^3}{10q^2}\binom{q+1}{\tau}$$

for large enough q. Since $|F_q^*| = m|A|$, by inclusion-exclusion,

$$\begin{aligned} N &\geq |F_q^*| \cdot \binom{q-2}{\tau-3} - \binom{|F_q^*|}{2}\binom{q-4}{\tau-5} \\ &= m|A| \cdot \binom{q+1}{\tau}\frac{\tau(\tau-1)(\tau-2)}{(q+1)q(q-1)}\Big(1 - \frac{(m|A|-1)(\tau-3)(\tau-4)}{2(q-2)(q-3)}\Big). \end{aligned}$$

Since q is large, $|A|\tau^2 \leq q$ and $m \leq (q+1)/6$, the second term in the brackets is at most 0.1. Also, since $\tau \geq 50$, we have $\tau(\tau-1)(\tau-2) \geq 0.94\tau^3$. Together with $m > q/6 - 1$, this gives the bound on N. $\square$

By a construction of Behrend, we can take

$$|A| = \frac{q}{e^{c\sqrt{\log q}}}$$

for some constant $c > 0$. As before, we construct H_q^* by placing a randomly permuted copy of F_q^* in each edge of G_q. The following lemma is a straightforward consequence of Lemma 14

Lemma 15. *Let q be large enough, $\tau \geq 50$ and $|A|\tau^2 \leq q$. Let I be a τ-element subset of $e \in E(G_q)$. Then the probability that I is independent in H_q^* is at most*

$$1 - \frac{|A|\tau^3}{10q^2}.$$

The expected number of independent sets of size $2\tau n/q$ in H_q^* is then at most

$$\Big(1 - \frac{|A|\tau^3}{10q^2}\Big)^{n-2n/\tau}\binom{n}{2\tau n/q}$$

using Lemma 15 and Corollary 12 as in the proof of Theorem 1. Taking

$$\tau = \Big\lfloor \sqrt{200q\log n/|A|}\Big\rfloor$$

the preceding quantity is $o(1)$ as $n \to \infty$, and so with high probability

$$\alpha(H_q^*) \le \frac{400n\sqrt{\log n}}{\sqrt{q|A|}}.$$

The construction of Behrend [3] guarantees that we can take $|A| = q/\exp(c\sqrt{\log n})$, for some constant $c > 0$, and therefore we obtain

$$\alpha(H_q^*) < \frac{n^{2/3}}{e^{c_1\sqrt{\log n}}}$$

for some constant $c_1 > 0$. This proves the lower bound in Theorem 3.

Connection to bounds on $r_3(N)$. The above argument shows

$$\alpha(H_q^*) = O\Big(\frac{n\sqrt{\log n}}{\sqrt{qr_3(q)}}\Big).$$

If for some constant $c > 0$, we are able to show

$$RL(C_3, K_t^3) = O\Big(\frac{t^{3/2}}{(\log t)^{c/3+3/4}}\Big),$$

then every n-vertex linear triangle-free triple system H has

$$\alpha(H) = \Omega(n^{2/3}(\log n)^{c/2+1/2}).$$

Applying this statement to H_q^*, we obtain

$$\frac{n\sqrt{\log n}}{\sqrt{qr_3(q)}} = \Omega\Big(n^{2/3}(\log n)^{c/2+1/2}\Big).$$

Since $q \sim n^{1/3}$ as $n \to \infty$, writing $\rho = r_3(q)/q$, we obtain

$$\rho = O\Big(\frac{1}{(\log n)^c}\Big).$$

This completes the verification that

$$r_3(N) = O\Big(\frac{N}{(\log N)^c}\Big).$$

4.3 Proof of Theorem 4

For Theorem 4, which states that

$$R(C_k, K_t^r) = \Omega^*\left(t^{1+\frac{1}{3k-1}}\right),$$

we let $G_{k,q}$ be an n-vertex $(q+1)$-uniform $(q+1)$-regular hypergraph with no cycles of length at most k, such that q is a maximum relative to n and such that $\lambda(G_{k,q}) \le 2\sqrt{q}$.

A construction of hypergraphs $G_{k,q}$ for primes $q \equiv 1 \bmod 4$ can be obtained from the construction of Ramanujan graphs of Lubotzsky, Phillips and Sarnak [13]. These $G_{k,q}$ are constructed from the following bipartite graphs of [13]: Let p, q be primes congruent to 1 modulo 4 with $p > 16$. If $(\frac{p}{q}) = -1$, then $B_{p,q}$ is a bipartite $(q+1)$-regular graph with $p(p^2-1)$ vertices in each part and no cycle of length less than $4\log_q(p/4)$. If $(\frac{p}{q}) = 1$, then $B_{p,q}$ is a bipartite $(q+1)$-regular graph with $p(p^2-1)/2$ vertices in each part and no cycle of length less than $2\log_q p$. In both cases $B_{p,q}$ has no cycle of length less than $2\log_q p$ since $p > 16$, and the second largest eigenvalue in absolute value except the first and last is at most $2\sqrt{q}$.
It follows that for each $k \ge 4$, $B_{p,q}$ is the bipartite incidence graph of a C_k-free $(q+1)$-graph $G_{k,q}$ on n vertices, where $n \in \{\frac{1}{2}p(p^2-1), p(p^2-1)\}$ as long as $2k < 2\log_q p$. The condition on k is equivalent to $q < p^{1/k}$, and in both cases, this is satisfied as long as $q < (1+o(1))(2n)^{1/3k}$. Furthermore, it follows that $\lambda(G_{k,q}) \le 2\sqrt{q}$.

Let $F_{q,r}$ denote the r-graph consisting of a union of $\tau = \lfloor 200 \log q \rfloor$ stars on q vertices, defined in the proof of Theorem 2. In each edge of $G_{k,q}$, put a randomly permuted copy of $F_{q,r}$ to get the r-graph $H_{k,q,r}$. Corollary 12 shows that if X is a set of at least $2\tau n/q$ vertices of $H_{k,q,r}$, then at least $n - 8n/\tau$ edges of $G_{k,q}$ contain at least τ vertices of X. By Lemma 13, the expected number of independent sets in $H_{k,q,r}$ of size $2\tau n/q$ is at most

$$\left(1 - \frac{\tau^2}{10q}\right)^{n-8n/\tau} \binom{n}{2\tau n/q} < \exp\left(-\frac{\tau^2 n}{20q} + \frac{2\tau n \log n}{q}\right)$$

provided q is large enough. The choice of τ ensures this decays to zero. Therefore with positive probability,

$$\alpha(H_{k,q,r}) = O\left(\frac{\tau n}{q}\right) = O\left(n^{1-1/3k} \log n\right),$$

as long as $q > c_k n^{1/3k}$ for some constant c_k depending only on k.

Now suppose we are given $k \ge 4$ and an integer n not of the form required to construct $B_{p,q}$ and hence $G_{k,q}$ and $H_{k,q,r}$. For such an n, we will choose p, q so that the construction above is possible on n' vertices with $n < n' < 8n$, and then restrict the resulting $H_{k,q,r}$ (which has n' vertices) to a subhypergraph with only n vertices. The resulting n-vertex r-graph would again have independence number $O\left(n^{1-1/3k} \log n\right)$.

Given $k \ge 4$ and a sufficiently large n, choose a prime $q \equiv 1 \bmod 4$ such that

$$\frac{1}{2}(2n)^{1/3k} < q < (2n)^{1/3k}.$$

Such a q exists by the prime number theorem in arithmetic progressions. Next choose a prime $p \equiv 1 \bmod 4$ such that

$$(3n)^{1/3} < p < 2n^{1/3}.$$

Again, by the prime number theorem in arithmetic progressions, we can find such a p because n is sufficiently large. Now set $n' = p(p^2-1)/2$ or $p(p^2-1)$ depending on whether $(\frac{p}{q})$ is 1 or -1, and construct $H_{k,q,r}$ as described above. The resulting $(q+1)$-graph $H_{k,q,r}$ contains no C_k as $q < (2n)^{1/3k} < (3n)^{1/3k} < p^{1/k}$. Finally, observe that

$$n' > p^3/2 - p/2 > 3n/2 - n^{1/3} > n$$

and $n' < p^3 < 8n$. Moreover, $q > c_k n^{1/3k}$ so the above bound on the independence number holds as $n \to \infty$.

This shows that for any $r \geq 3$ and $k \geq 4$,

$$R(C_k, K_t^r) = \Omega^*\left(t^{1+\frac{1}{3k-1}}\right).$$

4.4 Proof of Theorem 5

The specialization of the above arguments to $k = 5$ comes from the existence of generalized hexagons. These can be viewed as $(q+1)$-uniform $(q+1)$-regular hypergraphs G_q on $q^5+q^4+q^3+q^2+q+1$ vertices containing no cycles of length at most five, and moreover the associated matrix $A(G_q)$ has $\lambda(G_q) = \sqrt{q}$ once more. Using the hypergraph $F_{q,r}$ in each edge of the hypergraph G_q as before gives the result: we obtain a hypergraph $H_{5,q,r}$ with

$$\alpha(H_{5,q,r}) = O(n^{4/5}\log n)$$

from which the lower bound on Ramsey numbers $R(C_5, K_t^r) = \Omega^*(t^{5/4})$ for all $r \geq 3$ follows.

References

[1] Ajtai, M., Komlós, J., Pintz, J., Spencer, J., and Szemerédi, E. A note on Ramsey numbers, J. Comb. Theory (Series A), 29, 354-360, (1980).

[2] Ajtai, M., Komlós, J., Szemerédi, E., A note on Ramsey numbers. J. Comb. Theory (Series A), 29, 354-360, (1980).

[3] Behrend, F., On sets of integers which contain no three elements in arithmetic progression, Proc. Nat. Acad. Sci 32, 331–332, (1946).

[4] Benson, C, Minimal Regular Graphs of Girth 8 and 12. Canad. J. Math. 18, 1091-1094, (1966).

[5] Berge, C, Graphs and hypergraphs. Translated from the French by Edward Minieka. North-Holland Mathematical Library, Vol. 6. North-Holland Publishing Co., Amsterdam-London; American Elsevier Publishing Co., Inc., New York, 1973. xiv+528 pp.

[6] Bohman, T., Keevash, P., The early evolution of the H-free process, Invent. Math, 181 no. 2 291–336, (2010).

[7] Duke, R., Lefmann, H., Rödl, V., On uncrowded hypergraphs, Random Structures & Algorithhms, 6, 209–212, (1995).

[8] Haemers, W., Interlacing eigenvalues and graphs, Linear algebra Appl. 226/228, 593–616, (1995).

[9] Keevash, P, Sudakov, B, Verstraete, J. Bipartite Turán Numbers and a conjecture of Erdős and Simonovits. Preprint (2011).

[10] Kim, J., The Ramsey number $R(3,t)$ has order of magnitude $t^2/\log t$. Random Structures & Algorithms 7, 173-207, (1995).

[11] Kostochka, A., Rödl, V., Constructions of sparse uniform hypergraphs with high chromatic number, Random Structures Algorithms 36 (2010), 4656.

[12] Lovász, L. On chromatic number of finite set-systems. Acta Math. Acad. Sci. Hungar. 19 1968 5967.

[13] Lubotzky, A., Phillips, R., Sarnak, P., Ramanujan Graphs, Combinatorica 8 (3), 261–277, (1988).

[14] Margulis, G., Explicit constructions of graphs without short cycles and low density codes, Combinatorica 2 (1), 71–78, (1982).

[15] Ruzsa, I., Szemerédi, E. Triple systems with no six points carrying three triangles, in Combinatorics, Keszthely, 1976, Coll. Math. Soc. J. Bolyai 18 Volume II., 939–945.

[16] Sanders, T., On Roth's theorem on progressions, Annals of Mathematics, to appear

[17] Tits, J. Sur la trialité et certains groupes qui s'en d'eduisent. Publ. Math. I.H.E.S. Paris 2, 14–60, (1959)

[18] Tomescu, I, Sur le probléme du coloriage des graphes généralisés. (French) C. R. Acad. Sci. Paris Sér. A-B 267 1968 A250 – A252.